Experiment 7: Managing Data and Models using DVC with Cloud Remote

---

AIM

To study data and model versioning using DVC integrated with Git and cloud storage.

OBJECTIVES

1. To understand the importance of data and model version control in ML projects.
2. To study offline setup of DVC for local tracking.
3. To understand remote storage integration using Azure or GCP.

THEORY

1. Data and Model Versioning

In ML projects, datasets and models change frequently. Git tracks code efficiently but is not suitable for large data files.

DVC (Data Version Control) is used to manage large datasets and model files along with Git. It stores metadata in Git while keeping actual data separately, ensuring reproducibility and organized workflows.

---

2. Offline Setup: DVC with Git

In the offline setup, DVC is initialized inside a Git repository to track datasets and model files locally.

DVC maintains lightweight tracking files, allowing users to switch between different data or model versions easily without storing large files directly in Git.

---

3. Cloud Remote Storage

For collaboration, DVC supports remote storage systems such as Azure Blob Storage or GCP Cloud Storage.

The actual data is stored in the cloud, while version information remains in Git. This enables secure sharing and efficient data management across teams.

---

4. Integrated Workflow

In practice, Git manages code, and DVC manages data and models. Locally, changes are tracked using DVC, and remotely, data is stored in cloud storage.

This setup ensures scalable, reproducible, and collaborative ML project management.

CONCLUSION

In this experiment, we studied data and model management using DVC with Git and cloud storage integration.

This approach improves reproducibility, collaboration, and efficient handling of large ML artifacts.